

- 1 The photograph shows an aquatic plant, *Elodea canadensis*.



(Source: © Mikko Suonio/Alamy Stock Photo)

Magnification $\times 1$

These plants were originally from North America but are now found in Europe, South America, Asia, Australasia and parts of Africa. They live in ponds, lakes and slow-flowing rivers.

If the stem of the plant is cut underwater, bubbles of oxygen gas are released from the cut end of the stem as the plant photosynthesises.

(a) Describe an experiment to investigate the effect of different wavelengths of light on the rate of photosynthesis in *Elodea canadensis*.

(5)

(b) Explain how the features of the grana in a chloroplast enable photosynthesis to occur.

(3)

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(Total for Question 1 = 8 marks)
